

IT342 Phase 1 Development – Google Sign-In Authentication Integration

Project: Knock Knock

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Course: IT342

1. GitHub Repository Link

Repository URL: https://github.com/theninanicole/IT342_KnockKnock_G4_Villadarez

2. Brief Description of Google Login Implementation

Google authentication was integrated as an alternative to email/password login and registration. Users can now sign in or sign up using their Google account from the existing login and register pages. The system supports both regular authentication and Google-based authentication in one unified auth module.

Backend Technologies/Libraries Used

- Spring Boot 3 (REST API + Security configuration)
- Spring Security (route protection and authentication flow)
- JWT for session/token issuance
- Google token verification using google-api-client (GoogleIdTokenVerifier)
- JPA/Hibernate for persistence (user storage, including google_id and auth provider)

Frontend Integration Summary (ReactJS)

- Added Google sign-in flow using Google Identity Services script
- Connected Google credential flow to backend endpoints:
 - POST /api/auth/oauth/google
 - POST /api/auth/register/visitor/google
 - POST /api/auth/register/condo-admin/google
- Kept existing Google button UI and styling unchanged
- Added redirect logic after successful Google auth based on role:
 - Visitor → Visitor Dashboard

- Condo Admin → Admin Dashboard
- Added user feedback for failed auth cases (invalid token, provider conflict, etc.)

Backend Feature Coverage

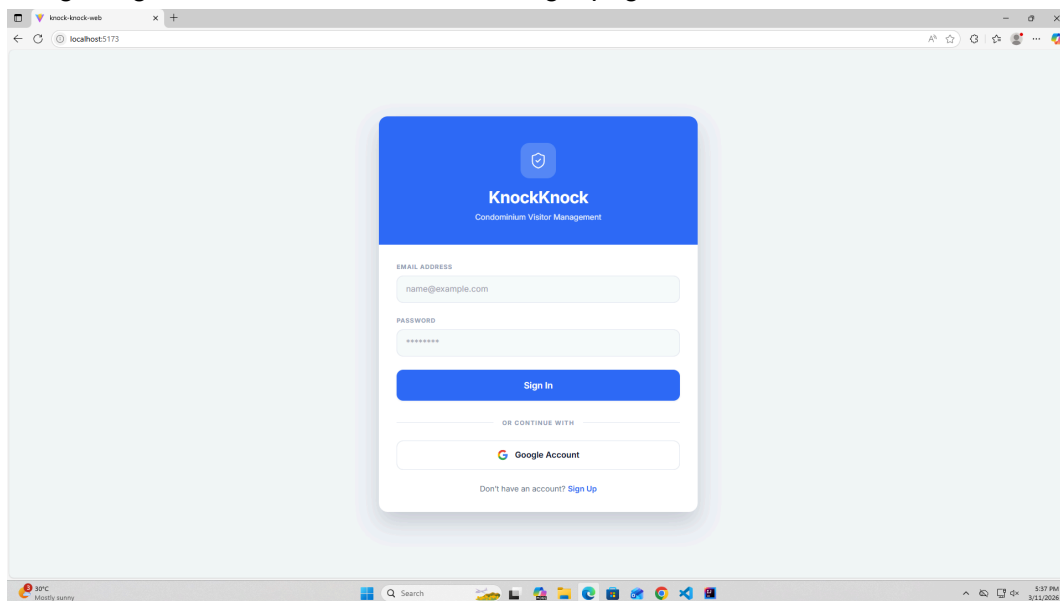
- Google-authenticated users are validated and processed in backend
- Existing users with email/password are protected from provider conflict (proper error response)
- Google users are saved/updated with:
 - internally generated password_hash
 - authProvider = google
 - google_id value
- Existing login (/auth/login) still works for email/password users
- Security routing remains protected and consistent with current module

Challenges Encountered

- Origin mismatch and Google Console configuration (origin not allowed)
- Browser popup/COOP behavior during local development (localhost)

3. Proof of Implementation

Google Sign-in button in the frontend in login page



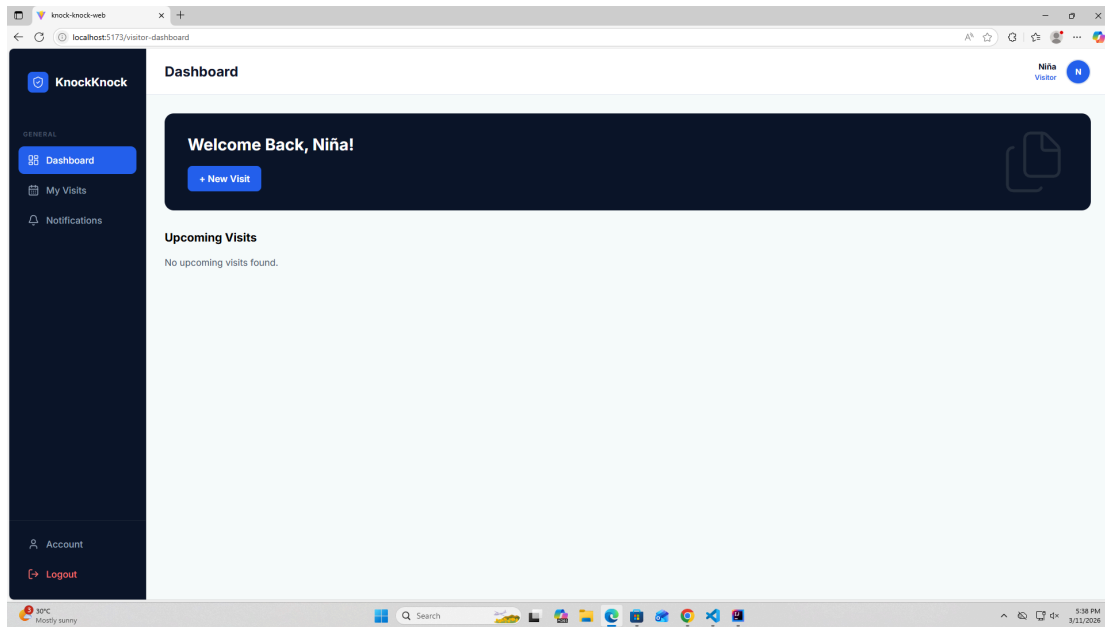
Google Sign-up button in the frontend in registration page

The screenshot shows a web browser window with the URL `localhost:5173/register`. The page features a registration form for 'KnockKnock' with the subtitle 'Create your account to get started'. The form has two tabs: 'Visitor' (selected) and 'Administrator'. Under the 'PERSONAL INFORMATION' section, there are input fields for 'FULL NAME' (containing 'Juan Dela Cruz'), 'EMAIL ADDRESS' (containing 'name@email.com'), 'CONTACT NUMBER' (containing '09123456789'), 'PASSWORD', and 'CONFIRM PASSWORD'. A blue 'Create Account' button is positioned below these fields. Below the button, there is a section 'OR CONTINUE WITH' featuring a 'Google Account' button. At the bottom, a link 'Already have an account? Sign In' is visible. The browser's taskbar at the bottom shows a temperature of 30°C and the date 3/11/2026.

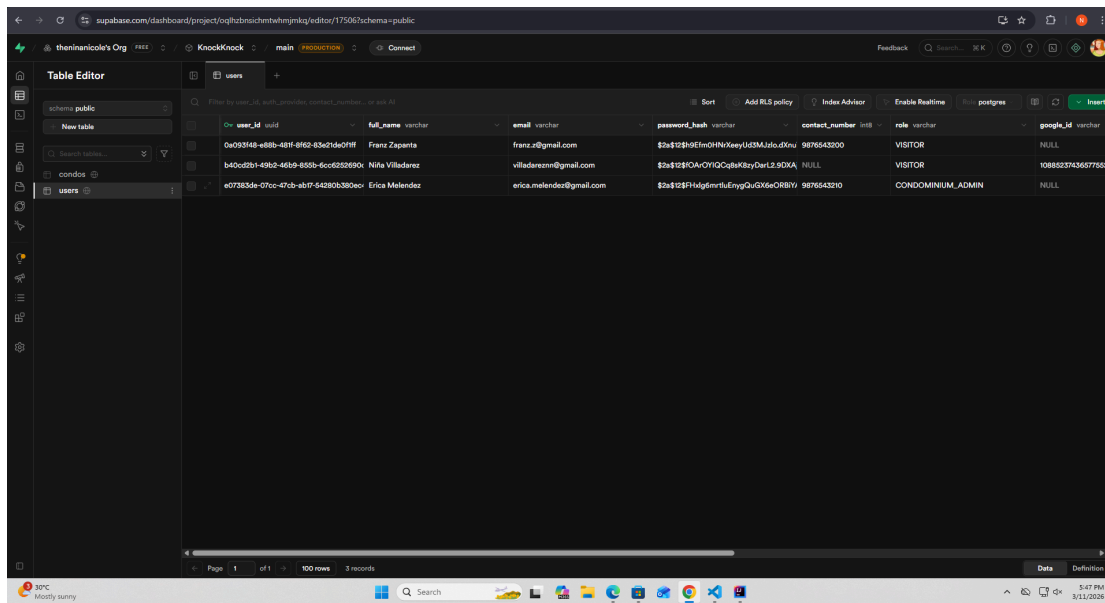
Google account selection or authentication page

This screenshot shows the same 'KnockKnock' login page, but with a Google account selection window open in the foreground. The window, titled 'Sign in - Google accounts - Profile 1 - Microsoft Edge', displays the URL `https://accounts.google.com/v3/signin/accountchoosertas=0MeHX...` and the text 'Sign in with Google'. It prompts the user to 'Choose an account to continue to KnockKnock' and lists two accounts: 'Niña' (nwilladarez18@gmail.com) and 'Niña Villadarez' (villadarezm@gmail.com). There is also an option to 'Use another account'. The background login form is partially visible, showing fields for 'EMAIL ADDRESS' (containing 'name@example.com') and 'PASSWORD', along with a 'Sign In' button and a 'Google Account' button. The 'Don't have an account? Sign Up' link is also present. The browser's taskbar at the bottom shows a temperature of 30°C and the date 3/11/2026.

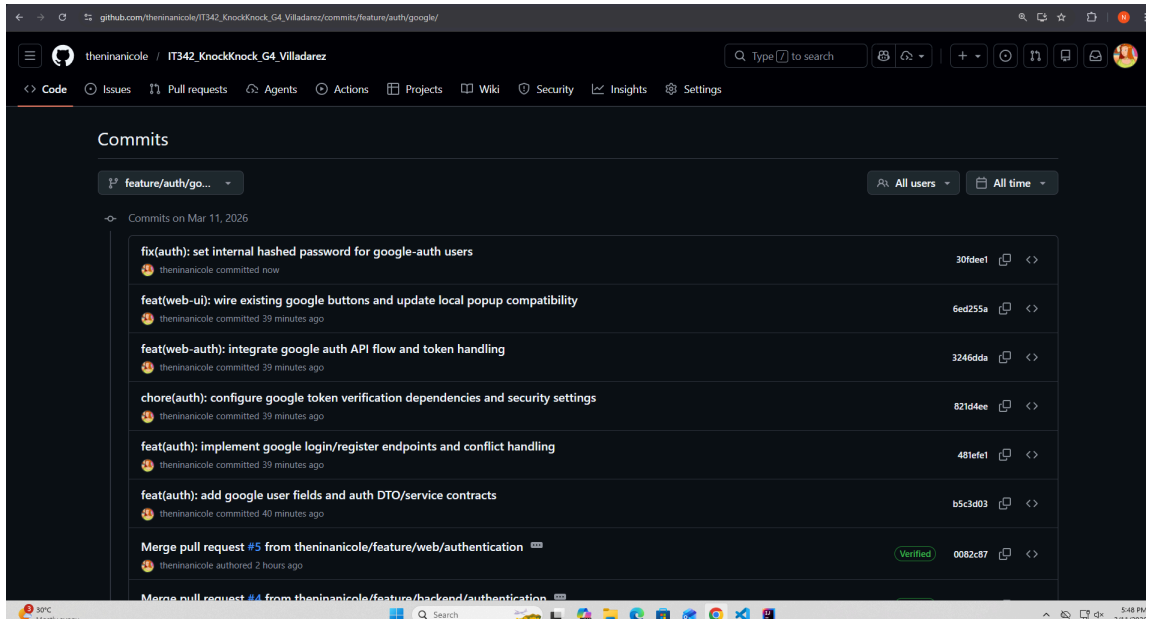
Successful login result in the system



Database



4. Commit History Evidence



The screenshot displays the GitHub commit history for the repository `theninanicole/IT342_KnockKnock_G4_Villadarez`. The page is filtered to show commits for the `feature/auth/google/` branch. The commit history is sorted by time, showing the most recent commits first. The commits are as follows:

Commit Message	Author	Time	Commit Hash	Actions
<code>fix(auth): set internal hashed password for google-auth users</code>	theninanicole	committed now	30fdee1	View Compare
<code>feat(web-ui): wire existing google buttons and update local popup compatibility</code>	theninanicole	committed 39 minutes ago	6ed255a	View Compare
<code>feat(web-auth): integrate google auth API flow and token handling</code>	theninanicole	committed 39 minutes ago	3246dda	View Compare
<code>chore(auth): configure google token verification dependencies and security settings</code>	theninanicole	committed 39 minutes ago	821d4ee	View Compare
<code>feat(auth): implement google login/register endpoints and conflict handling</code>	theninanicole	committed 39 minutes ago	481efe1	View Compare
<code>feat(auth): add google user fields and auth DTO/service contracts</code>	theninanicole	committed 40 minutes ago	b5c3d03	View Compare
Merge pull request #5 from theninanicole/feature/web/authentication	theninanicole	authorized 2 hours ago	0082c87	View Compare
Merge pull request #4 from theninanicole/feature/backend/authentication	theninanicole	authorized 2 hours ago	0082c87	View Compare